

CALCULUS (II)

WEEK (XV)

VECTOR FIELDS

Definition A function whose domain and range are subsets of Euclidean 3-space, $\mathbb{R}^3$, is called a vector field. Thus a vector field F associates a vector $F(x, y, z)$ with each point (x, y, z) in its domain. $F(x, y, z)$ can be expressed in terms of the standard basis in $\mathbb{R}^3$ as

$$F(x, y, z) = F_1(x, y, z)\vec{i} + F_2(x, y, z)\vec{j} + F_3(x, y, z)\vec{k}$$

$F_1(x, y, z), F_2(x, y, z), F_3(x, y, z)$: scalar-valued (real valued) functions
that is $F_i: \mathbb{R}^3 \rightarrow \mathbb{R} \quad i=1, 2, 3$
 $(x, y, z) \rightarrow F_i(x, y, z)$

$F(x, y) = F_1(x, y)\vec{i} + F_2(x, y)\vec{j}$ is plane vector field
or vector field in xy -plane.

Ex $\nearrow$ Velocity vector fields (wind, ocean currents)
Force fields (gravitational force)

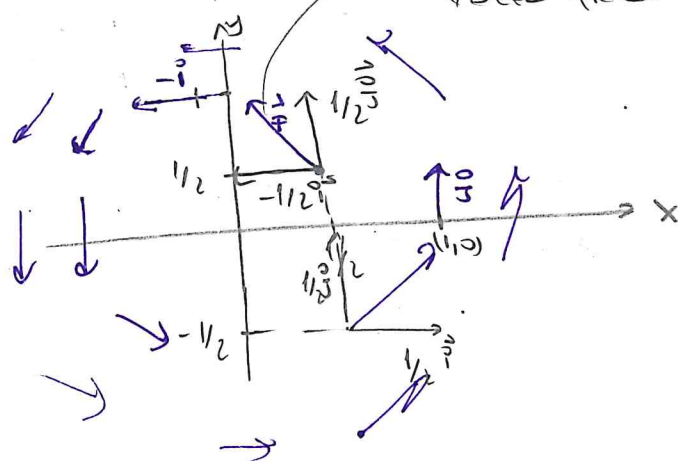
Ex $\nearrow$ Sketch the vector field $\vec{F}(x, y) = -y\vec{i} + x\vec{j}$ at point $(1/2, 1/2)$ vector field.

$$\vec{F}(1/2, 1/2) = -1/2\vec{i} + 1/2\vec{j}$$

$$\vec{F}(1/2, -1/2) = 1/2\vec{i} + 1/2\vec{j}$$

$$\vec{F}(1, 0) = \vec{j}$$

$$\vec{F}(0, 1) = -\vec{i}$$



→ Notice that we have already seen a vector field function.

$$\nabla f = \text{grad } f = \langle f_x, f_y, f_z \rangle \text{ gradient vector-field.}$$

(in three dimensional space)

Ex Find gradient vector field of $f(x, y, z) = z \cdot e^{-xy}$

$$\nabla f = \text{grad } f = \langle -yz e^{-xy}, -ze^{-xy}, e^{-xy} \rangle$$

FIELD LINES (INTEGRAL CURVES, TRAJECTORIES, STREAMLINES)

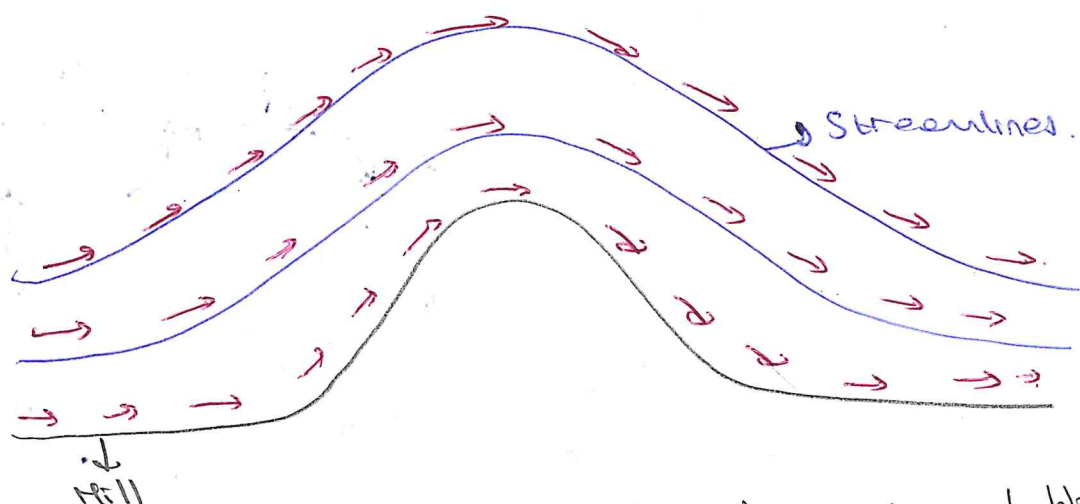
Question What path will be followed by a particle whose velocity is given by a vector field.

Answer The path will be a curve to which the field is tangent at every point

→ Such curves are called field lines, integral curves or trajectories for the given vector field.

→ In specific case where the vector field gives the velocity in a fluid flow, the field lines are also called streamlines or flow lines of the flow.

→ For a force field, the field lines are called lines of force



The velocity field and some streamlines of wind blowing over a hill (7)

→ The field lines of F do not depend on the magnitude of F at any point but only on the direction of the field.

If the field line through some point has parametric equation $\vec{r} = \vec{r}(t)$, then its tangent vector $d\vec{r}/dt$ must be parallel to $F(\vec{r}(t))$ for all t . Thus,

$$\frac{d\vec{r}}{dt} = \lambda(t) F(\vec{r}(t))$$

→ If we break the equation into components

$$\frac{dx}{dt} = \lambda(t) F_1(x, y, z) \quad \frac{dy}{dt} = \lambda(t) F_2(x, y, z) \quad \frac{dz}{dt} = \lambda(t) F_3(x, y, z)$$

→ Then, we can obtain equivalent differential expressions for $\lambda(t)dt$ and hence write

$$\frac{dx}{F_1(x, y, z)} = \frac{dy}{F_2(x, y, z)} = \frac{dz}{F_3(x, y, z)}$$

Ex 7 Find field lines of

$$F(x, y, z) = -km \frac{(x-x_0)\vec{i} + (y-y_0)\vec{j} + (z-z_0)\vec{k}}{((x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2)^{3/2}}$$

Let $\gamma = -\frac{km}{((x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2)^{3/2}} \in \mathbb{R}$. The vector in the numerator gives direction therefore $\gamma \in \mathbb{R}$ doesn't have any importance.

$$\frac{dx}{(x-x_0)\gamma} = \frac{dy}{\gamma(y-y_0)} = \frac{dz}{\gamma(z-z_0)}$$

$$\frac{dx}{x-x_0} = \frac{dy}{y-y_0} = \frac{dz}{z-z_0}$$

$$\ln|x-x_0| + C_1 = \ln|y-y_0| + C_2 = \ln|z-z_0| + C_3$$

$$C_1(x-x_0) = C_2(y-y_0) = C_3(z-z_0)$$

Straight lines through

(2)

Ex 7 Find the field lines of the velocity field $V = \sqrt{2}(-y\vec{i} + x\vec{j})$.

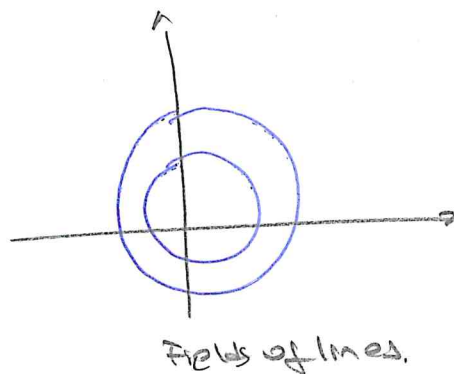
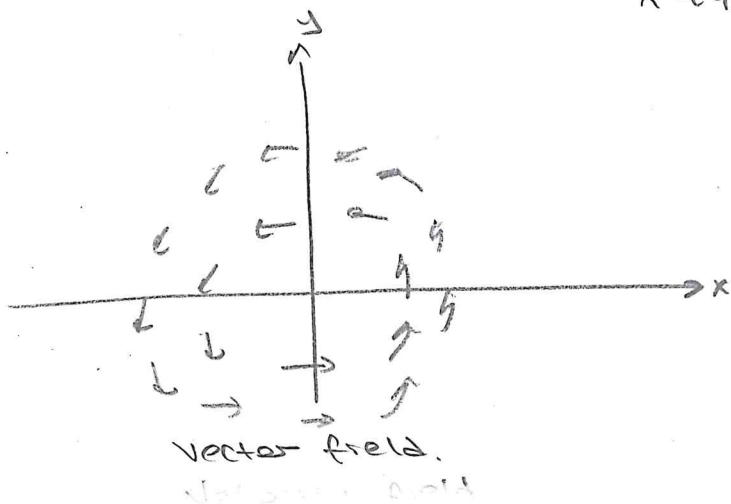
$$\frac{dx}{-y} = \frac{dy}{x}$$

$$\int x dx = \int -y dy$$

$$\frac{x^2}{2} + C_1 = -\frac{y^2}{2} + C_2$$

$$x^2 + y^2 = 2(C_2 - C_1)$$

$$x^2 + y^2 = C$$



Ex 7 Find field lines of $F = xz\vec{i} + 2xz^2\vec{j} + x^2\vec{k}$

$$\frac{dx}{xz} = \frac{dy}{2xz^2} = \frac{dz}{x^2}$$

$$dy = 2x dx \quad \text{and} \quad dy = 2z dz$$

$$y = x^2 + C_1$$

$$y = z^2 + C_2$$

$$\left. \begin{array}{l} y = x^2 + C_1 \\ y = z^2 + C_2 \end{array} \right\} \begin{array}{l} \text{parabolic} \\ \text{cylinders} \end{array}$$

The field lines are the curves of intersection of two families

of parabolic cylinders $y = x^2 + C_1$ and $y = z^2 + C_2$.

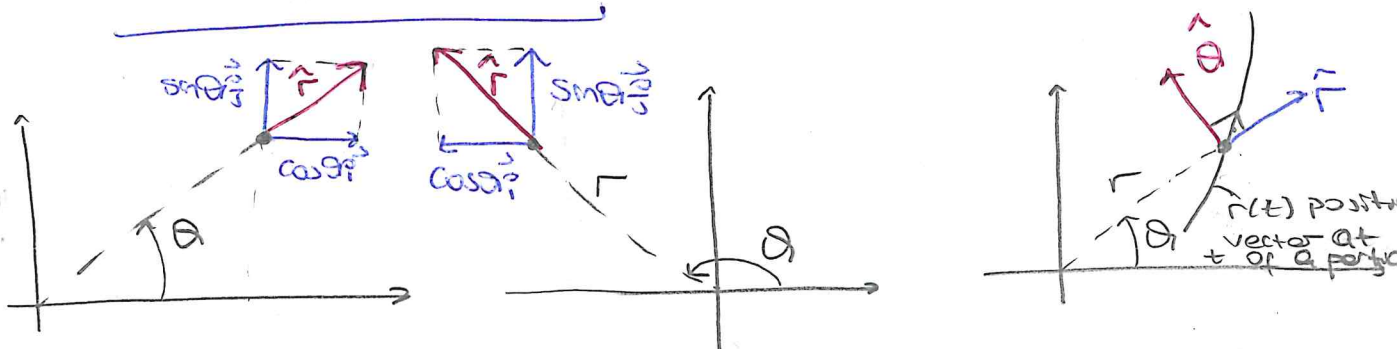
VECTOR FIELDS IN SCALAR COORDINATES

→ A vector field in the plane can be expressed in terms of polar coordinates in the form $\boxed{\vec{F} = F(r, \theta) = F_r(r, \theta) \hat{r} + F_\theta(r, \theta) \hat{\theta}}$

where $\hat{r}$ and $\hat{\theta}$, defined everywhere except at the origin by

$$\begin{aligned}\hat{r} &= \cos\theta \vec{i} + \sin\theta \vec{j} \\ \hat{\theta} &= -\sin\theta \vec{i} + \cos\theta \vec{j}\end{aligned}$$

are unit vectors in the direction of increasing r at (r, θ) .



Here, $F_r(r, \theta)$ is called the **radial component** of F .
 $F_\theta(r, \theta)$ is called the **transverse component** of F .

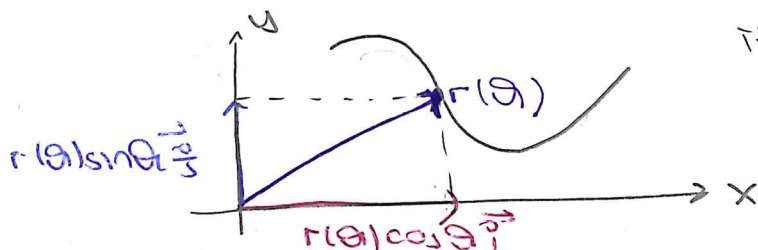
Note that $\frac{d\hat{r}}{d\theta} = -\sin\theta \vec{i} + \cos\theta \vec{j} = \hat{\theta}$

$$\boxed{\frac{d\hat{r}}{d\theta} = \hat{\theta} \quad \frac{d\hat{\theta}}{d\theta} = -\hat{r}}$$

And also $\hat{\theta}$ is just rotated $\hat{r}$ rotated 90° counterclockwise

→ A curve with polar equation $r = r(\theta)$ can be expressed in vector parametric form

$$\vec{r} = r \hat{r}$$



This curve is a field line of F
 it is differential tangent vector

$$d\vec{r} = dr \hat{r} + r \frac{d\hat{r}}{d\theta} d\theta$$

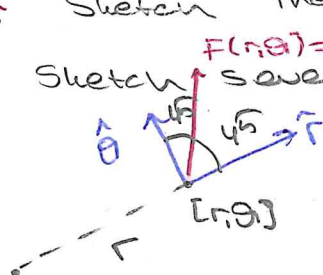
$$\boxed{d\vec{r} = \hat{r} dr + r \hat{\theta} d\theta}$$

is parallel to the vector field $F(r, \theta)$ at any point except origin, that is, if $r = f(\theta)$ satisfies the differential equation

$$\frac{dr}{F_r(r, \theta)} = \frac{r d\theta}{F_\theta(r, \theta)}$$

Ex Sketch the vector field $F(r, \theta) = \hat{r} + \hat{\theta}$ and find its field lines.

Sketch several field lines.



Choosing $\alpha = 45^\circ$ with $\hat{r}$

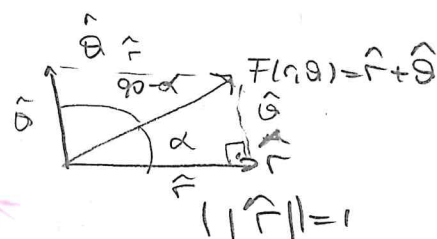
$$F_r(r, \theta) = F_\theta(r, \theta) = 1$$

$$dr = r d\theta \Rightarrow \frac{dr}{r} = d\theta$$

$$\frac{F_r(r, \theta)}{F_\theta(r, \theta)} = \frac{1}{1} \Rightarrow \frac{dr}{r} = d\theta$$

$$\frac{dr}{r} = d\theta \Rightarrow \ln r = \theta + C$$

$$r(\theta) = e^\theta \cdot \frac{e^C}{C}$$



$$\|\hat{r}\| = 1$$

$$\|\hat{\theta}\| = 1$$

$$\alpha = 45^\circ$$

GRADIENT DIVERGENCE AND CURL

→ First-Order information about the rate of change of a three-dimensional scalar field, $f(x, y, z)$, is contained in the three-first partial derivatives.

$$\text{grad } f(x, y, z) = \nabla f(x, y, z) = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k}$$

collects this information into a single vector-valued "derivative" of f .

$$F(x, y, z) = F_1(x, y, z) \hat{i} + F_2(x, y, z) \hat{j} + F_3(x, y, z) \hat{k}$$

Divergence and curl

Consider we have a vector field.

$$F(x, y, z) = F_1(x, y, z)\vec{i} + F_2(x, y, z)\vec{j} + F_3(x, y, z)\vec{k}$$

$$\begin{aligned}\text{div } F &= \nabla \cdot F = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \\ &= \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}\end{aligned}$$

$$\nabla = \vec{i} \frac{\partial}{\partial x} + \vec{j} \frac{\partial}{\partial y} + \vec{k} \frac{\partial}{\partial z}$$

$$\begin{aligned}\text{curl } F &= \nabla \times F = \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right) \vec{i} + \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \right) \vec{j} + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \vec{k} \\ &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}\end{aligned}$$

Ex Find the divergence and curl of the vector fields.

$$F = xy\vec{i} + (y^2 - z^2)\vec{j} + yz\vec{k}$$

$$\text{div } F = \nabla \cdot F = \frac{\partial}{\partial x}(xy) + \frac{\partial}{\partial y}(y^2 - z^2) + \frac{\partial}{\partial z}(yz) = y + 2y + y = 4y.$$

$$\text{curl } F = \nabla \times F = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy & y^2 - z^2 & yz \end{vmatrix}$$

$$= \vec{i} \left(\frac{\partial}{\partial y}(yz) - \frac{\partial}{\partial z}(y^2 - z^2) \right)$$

$$- \vec{j} \left(\frac{\partial}{\partial x}(yz) - \frac{\partial}{\partial z}(xy) \right)$$

$$+ \vec{k} \left(\frac{\partial}{\partial x}(y^2 - z^2) - \frac{\partial}{\partial y}(xy) \right)$$

$$= \vec{i} (z - (-2z)) - \vec{j} \cdot 0 + \vec{k} (-x)$$

$$= \vec{i} (3z) + \vec{k} (-x)$$

For two dimensional vector field $F(x,y) = F_1(x,y)\vec{i} + F_2(x,y)\vec{j}$

then we have

$$\text{div } F = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} \quad \text{curl } F = \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \vec{k}$$

Ex Find divergence and curl of $F = xe^{xy}\vec{i} - ye^{xy}\vec{j}$

$$\text{div } F = \nabla \cdot F = \frac{\partial}{\partial x}(xe^y) + \frac{\partial}{\partial y}(-ye^x) = e^y - e^x$$

$$\text{curl } F = \nabla \times F = \left(\frac{\partial}{\partial x}(-ye^x) - \frac{\partial}{\partial y}(xe^y) \right) \vec{k} = -(e^x + e^y)\vec{k}$$

CONSERVATIVE VECTOR FIELDS

$$\rightarrow F(x,y,z) = \nabla \phi(x,y,z) = \frac{\partial \phi}{\partial x}\vec{i} + \frac{\partial \phi}{\partial y}\vec{j} + \frac{\partial \phi}{\partial z}\vec{k}$$

Gradient is a vector field.

$\rightarrow$ What about the converse?

Definition If $F(x,y,z) = \nabla \phi(x,y,z)$ in a domain D , then we say that F is **conservative** vector field in D , we call the function ϕ a **scalar (potential) function** of F on D .

Ex Show that $V = -\gamma y\vec{i} + \gamma x\vec{j}$ is not conservative if $\gamma \neq 0$

$$\frac{\partial \phi}{\partial x} = -\gamma y$$

$$\frac{\partial \phi}{\partial y} = \gamma x$$

$$\phi(x,y) = -\gamma xy + C_1(y)$$

$$\phi(x,y) = \gamma xy + C_2(x)$$

$$-\gamma xy + C_1(y) = \gamma xy + C_2(x)$$

$$2\gamma xy = C_1(y) - C_2(x) \text{ for all } (x,y)$$

IMPOSSIBLE

METHOD II Alternatively, if v has a potential ϕ , then we can form the mixed partial derivatives ϕ from the two equations and get

$$\frac{\partial^2 \phi}{\partial y \partial x} = -\gamma \quad \neq \quad \frac{\partial^2 \phi}{\partial x \partial y} = \gamma$$

This is not possible for $\gamma \neq 0$

Necessary condition for a conservative plane vector field.

If $F(x,y) = F_1(x,y)\vec{i} + F_2(x,y)\vec{j}$ is a conservative field in a domain D of the xy -plane, then the condition

$$\boxed{\frac{\partial F_1(x,y)}{\partial y} = \frac{\partial F_2(x,y)}{\partial x}} \text{ must be satisfied for all points of } D.$$

Necessary conditions for a conservative vector field in 3-space

If $F(x,y,z) = F_1(x,y,z)\vec{i} + F_2(x,y,z)\vec{j} + F_3(x,y,z)\vec{k}$ is a conservative vector field in a domain D in 3-space, then we must have

$$\boxed{\frac{\partial F_1}{\partial y} = \frac{\partial F_2}{\partial x} \quad \frac{\partial F_1}{\partial z} = \frac{\partial F_3}{\partial x} \quad \frac{\partial F_2}{\partial z} = \frac{\partial F_3}{\partial y}}$$

DEFINITION If $\phi(x,y,z)$ is a potential function for the conservative vector field F , then the level surfaces $\phi(x,y,z) = C$ of ϕ are called equipotential surfaces of F

→ Similarly, for a conservative plane vector field, the level curves of the potential function are called equipotential curves of the vector field. They are orthogonal trajectories of the field lines; that is they intersect the field lines at right angles

Ex Show that the vector field $F(x,y) = x\vec{i} - y\vec{j}$ is conservative and find a potential function.

CHECK CONSERVATIVITY

$$\frac{\partial \phi}{\partial x \partial y} = 0 \quad \frac{\partial \phi}{\partial y \partial x} = 0$$

$0 = 0$ so it is conservative

Find METHOD I

$$\frac{\partial \phi}{\partial x} = F_1 = x \quad \text{and} \quad \frac{\partial \phi}{\partial y} = F_2 = -y$$

$$\phi(x,y) = \int x dx + C_1(y)$$

$$\frac{x^2}{2} + C_1(y) = \phi(x,y) \quad \text{OR}$$

$$\frac{\partial \phi}{\partial y} = -y = C_1'(y)$$

$$C_1(y) = -\frac{y^2}{2} + C \quad C \in \mathbb{R}$$

METHOD II

$$\frac{\partial \phi}{\partial y} = -y$$

$$\phi(x,y) = -\frac{y^2}{2} + C_1(x)$$

$$\frac{\partial \phi}{\partial x} = C_1'(x) = x$$

$$C_1(x) = \frac{x^2}{2}$$

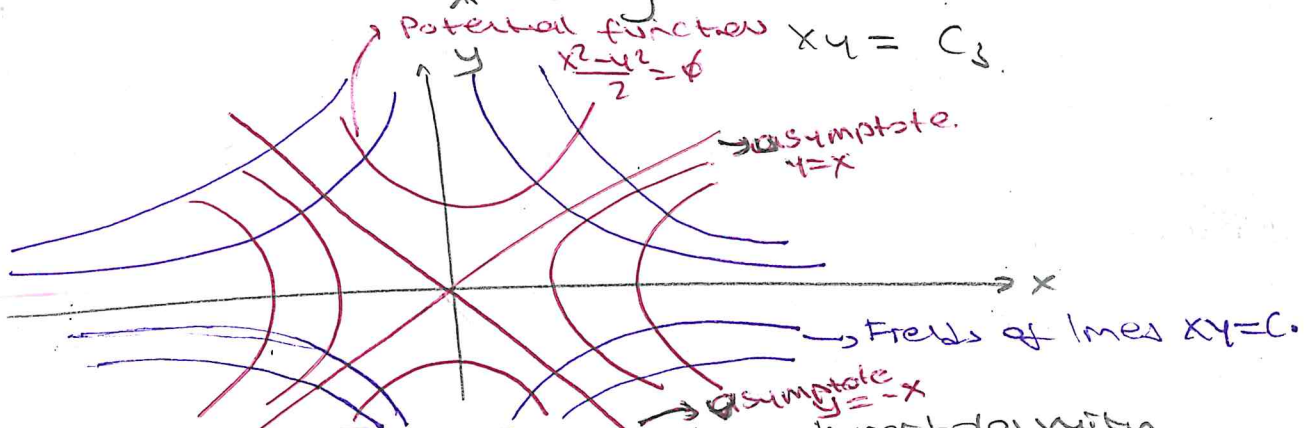
$\phi(x,y) = \frac{x^2 - y^2}{2} + C$ BOTH IS SAME $\phi(x,y) = \frac{x^2 - y^2}{2} + C$

Field lines of F

$$\frac{dx}{F_1} = \frac{dy}{F_2}$$

$$\frac{dx}{x} = -\frac{dy}{y}$$

$$\ln|x| = -\ln|y| + \ln C_3$$



→ The potential curves of F rectangular hyperbolas with coordinate axes as asymptote. (red lines)

→ $xy = C_3$ is a hyperbola for using transformation

$$u = \frac{x+y}{\sqrt{2}} \quad v = \frac{x-y}{\sqrt{2}}$$

$$xy = \frac{u^2 - v^2}{2} = C \Rightarrow u^2 - v^2 = 2C$$

center

The field lines $xy = C_3$ are thus rectangular hyperbolas with the coordinate axes as asymptotes (purple ones)

Ex: Show that $\vec{F} = \langle 3x^2, 2yz, y^2 \rangle$ is conservative and find its potential function.

$$\vec{F} = F_1 \vec{i} + F_2 \vec{j} + F_3 \vec{k} = \nabla \phi$$

$$F_1 = \frac{\partial \phi}{\partial x} \quad F_2 = \frac{\partial \phi}{\partial y} \quad F_3 = \frac{\partial \phi}{\partial z}$$

Check if such ϕ exists?

$$\frac{\partial F_1}{\partial y} = \frac{\partial F_2}{\partial x} \Rightarrow 0 = 0$$

$$\frac{\partial F_2}{\partial z} = \frac{\partial F_3}{\partial y} \Rightarrow 2y = 2y$$

$$\frac{\partial F_1}{\partial z} = \frac{\partial F_3}{\partial x} \Rightarrow 0 = 0$$

It is a conservative field.

Find its potential function, you can use three ways.

① $\phi(x, y, z) = \int F_1 dx + h(y, z)$ find $h(y, z)$ such that $\frac{\partial h}{\partial y} = F_2$ $\frac{\partial h}{\partial z} = F_3$

② $\phi(x, y, z) = \int F_2 dy + g(x, z)$ find $g(x, z)$ such that $\frac{\partial g}{\partial x} = F_1$ $\frac{\partial g}{\partial z} = F_3$

③ $\phi(x, y, z) = \int F_3 dz + k(x, y)$ find $k(x, y)$ such that $\frac{\partial k}{\partial x} = F_1$ and $\frac{\partial k}{\partial y} = F_2$

We will use ①

$$\phi = \int 3x^2 dx + h(y, z) = x^3 + h(y, z)$$

$$\frac{\partial \phi}{\partial y} = h'(y, z) = 2yz$$

$$h(y, z) = \int 2yz = y^2 z + u(z)$$

$$h'(z) = y^2 + u'(z) = y^2$$

$$\text{so } u'(z) = 0 \quad \text{so}$$

$$u(z) = C$$

$$\phi(x, y, z) = x^3 + y^2 z + C$$

Ex 7 Decide the following vector field is conservative

$$F = (xy - \sin z) \vec{i} + \left(\frac{1}{2}x^2 - \frac{e^y}{z}\right) \vec{j} + \left(\frac{e^y}{z^2} - x \cos z\right) \vec{k} \quad D = \{(x,y,z) : z \neq 0\}$$

If yes obtain its potential function.

$$\frac{\partial F_1}{\partial y} = x = \frac{\partial F_2}{\partial x}$$

$$\frac{\partial F_1}{\partial z} = -\cos z = \frac{\partial F_3}{\partial x}$$

$$\frac{\partial F_2}{\partial z} = \frac{e^y}{z^2} = \frac{\partial F_3}{\partial y}$$

Find ϕ $\phi(x,y,z) = \int (xy - \sin z) dx + h(y,z)$

$$= \left(\frac{x^2}{2}y - x \sin z\right) + h(y,z)$$

$$\frac{\partial \phi}{\partial y} = \frac{x^2}{2} + h_y(y,z) = \frac{x^2}{2} - \frac{e^y}{z}$$

$$h(y,z) = \int -\frac{e^y}{z} dy = -\frac{e^y}{z} + k(z)$$

$$\phi(x,y,z) = \frac{x^2}{2}y - x \sin z - \frac{e^y}{z} + k(z)$$

$$\frac{\partial \phi}{\partial z} = -x \cos z + \frac{e^y}{z^2} + k'(z) = \frac{e^y}{z^2} - x \cos z$$

$$k'(z) = 0 \quad k(z) = C$$

$$\phi(x,y,z) = \frac{x^2}{2}y - x \sin z - \frac{e^y}{z} + C$$

Ex 8 $F(x,y) \neq (0,0)$ $F(x,y) = \left(-\frac{y}{x^2+y^2}\right) \vec{i} + \left(\frac{x}{x^2+y^2}\right) \vec{j}$
 $\theta(x,y) = \text{polar angle } \theta \text{ of } (x,y) \text{ such that } 0 \leq \theta < 2\pi$

Then $x = r \cos(\theta(x,y))$ $y = r \sin(\theta(x,y))$

1) Show $\frac{\partial F_1}{\partial y} = \frac{\partial F_2}{\partial x}$? $\frac{\partial F_1}{\partial y} = \frac{\partial}{\partial y} \left(-\frac{y}{x^2+y^2}\right) = \frac{y^2-x^2}{(x^2+y^2)^2} = \frac{\partial}{\partial x} \left(\frac{x}{x^2+y^2}\right)$

2) Show $\nabla \phi = F$ for $0 < \theta < 2\pi$
 $x = r \cos \theta(x,y)$ $y = r \sin \theta(x,y)$

$$1 = \frac{\partial x}{\partial x} = \frac{\partial r}{\partial x} \cos \theta - r \sin \theta \frac{\partial \theta}{\partial x}$$

$$\frac{\partial \theta}{\partial x} = -\frac{r \sin \theta}{r^2} = -\frac{y}{x^2+y^2} = F_1$$

$$0 = \frac{\partial y}{\partial x} = \frac{\partial r}{\partial x} \sin \theta + r \cos \theta \frac{\partial \theta}{\partial x}$$

Similarly using $\frac{\partial x}{\partial y}, \frac{\partial y}{\partial y}$ gives $\frac{\partial \theta}{\partial y} = \frac{x}{x^2+y^2} = F_2$

These formulas hold for only $0 < \theta < 2\pi$; θ is not continuous on positive x axis. If $x > 0$ then $\lim_{+} \theta(x,y) = 0$ $\lim_{-} \theta(x,y) = 2\pi$

LINE INTEGRALS

- $\int_a^b f(x) dx$ total amount of quantity distributed along x axis between a and b .
- $\iint_D f(x,y) dA$, $\iiint_D f(x,y,z) dv$ total amounts of quantities distributed along the region D .
- $\int_C f(x,y,z) ds$ quantity distributed along a line C .

Definition

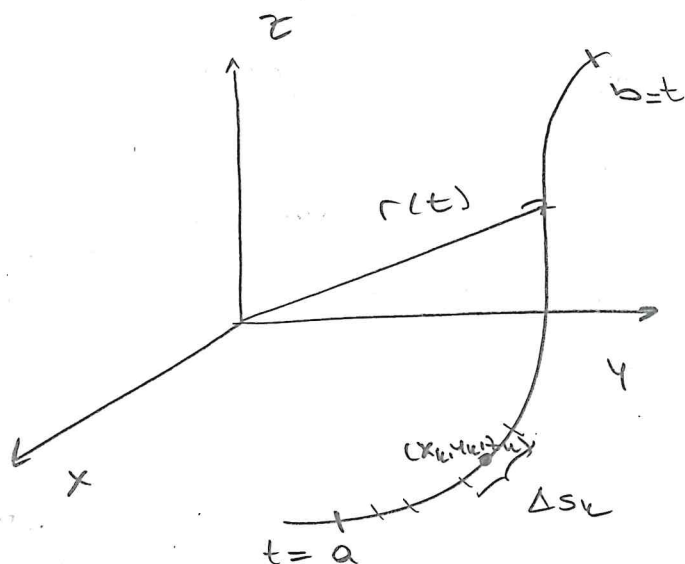
Let C be a bounded, continuous parametric curve in $\mathbb{R}^3$.

The curve C is a smooth curve if it has a parametrization of the form

$$\vec{r} = \vec{r}(t) = x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k} \quad t \in I$$

with "velocity" vector $\frac{d\vec{r}}{dt}$ continuous and nonzero

We will call C a smooth arc if it is a smooth curve with finite parameter interval $I = [a,b]$



Definition If f is defined on a curve C given parametrically by $\vec{r}(t) = p(t)\vec{i} + h(t)\vec{j} + k(t)\vec{k}$ $a \leq t \leq b$ then the line integral of f over C is

$$\int_C f(x,y,z) ds = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k, y_k, z_k) \Delta s_k$$

provided limit exists.

NOTE If C is a piecewise smooth such that
 $C = C_1 \cup C_2 \cup C_3 \dots \cup C_n$ then, $\int_C f ds = \int_{C_1} f ds + \int_{C_2} f ds + \dots + \int_{C_n} f ds$

→ Recall that for $\vec{r}(t) = x(t)\vec{i} + y(t)\vec{j}$ $a \leq t \leq b$ we have

$$\left| \frac{d\vec{r}}{dt} \right| = ds = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}, \text{ Length of } C = \int_C ds = \int_a^b \left| \frac{d\vec{r}}{dt} \right| dt$$

More generally line integrals evaluated similarly

$$\int_C f(x, y, t) ds = \int_a^b f(\vec{r}(t)) \left| \frac{d\vec{r}}{dt} \right| dt$$

NOTE Let we have different parametrization of
 Curve C such that $\vec{r}(t)$ $a \leq t \leq b$, and

Any point on $\vec{r}(t)$ on C can be expressed in terms of $\vec{r}^*(u)$ such
 that $u = u(t)$ $\alpha \leq u \leq \beta$, where $u(a) = \alpha$ and $u(b) = \beta$
 Then, we have

$$\int_a^b f(\vec{r}(t)) \left| \frac{d\vec{r}}{dt} \right| dt = \int_a^b f(\vec{r}^*(u(t))) \left| \frac{d\vec{r}^*}{du} \cdot \frac{du}{dt} \right| dt = \int_{\alpha}^{\beta} f(\vec{r}^*(u)) \left| \frac{d\vec{r}^*}{du} \right| du$$

$\alpha = u(a)$ $\beta = u(b)$

Thus, the line integral is independent of parametrization.
 of the curve C .

Ex 7 $\int_C y \, ds$ where $C: \vec{r} = a \cos t \vec{i} + a \sin t \vec{j} \quad (0 \leq t \leq \pi)$.

$$\frac{d\vec{r}}{dt} = -a \sin t \vec{i} + a \cos t \vec{j} \quad \left| \frac{d\vec{r}}{dt} \right| = \sqrt{a^2 \sin^2 t + a^2 \cos^2 t} = a$$

$$\int_0^\pi a \sin t \cdot a \, dt = -a^2 \cos t \Big|_0^\pi = 2a^2.$$

→ we can parametrize C also $\vec{r} = x \vec{i} + \sqrt{a^2 - x^2} \vec{j} \quad (-a \leq x \leq a)$

$$\frac{d\vec{r}}{dx} = \vec{i} - \frac{x}{\sqrt{a^2 - x^2}} \vec{j} \quad \left| \frac{d\vec{r}}{dx} \right| = \sqrt{1 + \frac{x^2}{a^2 - x^2}} = \frac{a}{\sqrt{a^2 - x^2}}$$

$$\int_{-a}^a \sqrt{a^2 - x^2} \cdot \frac{a}{\sqrt{a^2 - x^2}} \, dx = a \int_{-a}^a dx = 2a^2.$$

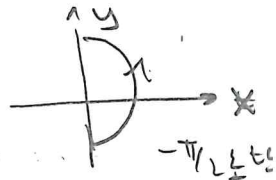
Ex 7 Evaluate $\int_C xy^4 \, ds$ where C is right half of the circle $x^2 + y^2 = 16$ oriented in the counterclockwise direction $-\pi/2 \leq t \leq \pi/2$.

• Parametrize the circle.

$$x = 4 \cos t$$

$$y = 4 \sin t$$

$$\vec{r}(t) = \langle 4 \cos t, 4 \sin t \rangle$$



$$\vec{r}'(t) = \langle -4 \sin t, 4 \cos t \rangle$$

$$ds = |\vec{r}'(t)| \, dt = \sqrt{16 \sin^2 t + 16 \cos^2 t} \, dt = 4 \, dt$$

$$\int_C xy^4 \, ds = \int_{-\pi/2}^{\pi/2} 4 \cos t (4 \sin t)^4 4 \, dt = 4096 \int_{-\pi/2}^{\pi/2} \cos t (\sin t)^4 \, dt$$

$$\sin t = u$$

$$\cos t \, dt = du$$

$$= 4096 \int_{-1}^1 u^4 \, du = \frac{4096}{5} (1 - (-1)) = \frac{8192}{5}$$

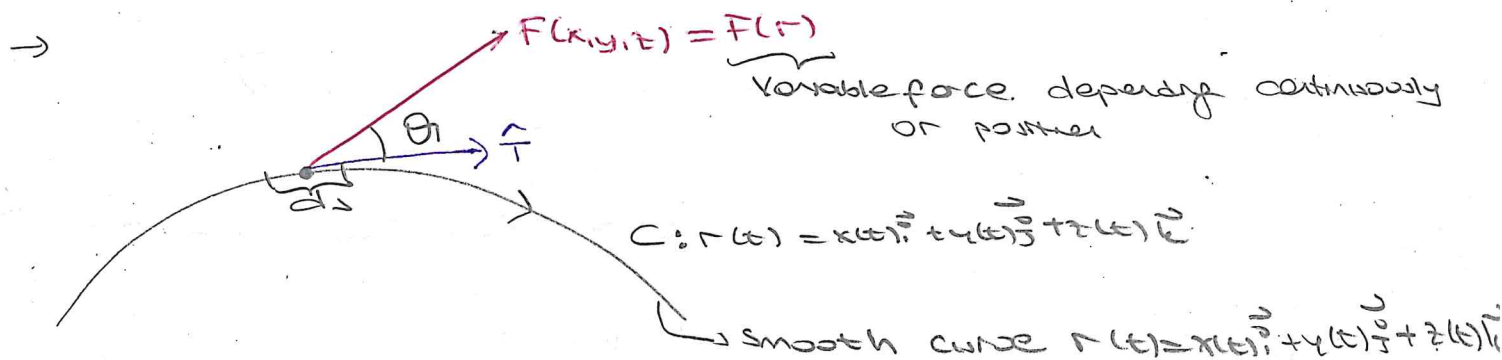
LINE INTEGRALS OF VECTOR FIELDS

→ In elementary physics

the work done by a constant force of magnitude F in moving an object a distance d $= F \cdot d$

(NOTE Here, the force is exerted in the direction of the motion)
upstream

→ If the object moves in a direction different from that of the force (because of some other forces acting on it), then the work done by the particular force is the product of the distance moved and the component of the force in the direction of motion.



→ The work done by $F(x, y, z) = F(r)$ in a moving object along a smooth curve C is the integral of work elements dW

$$dW = F(r) \cdot \hat{T} ds = F \cdot dr \quad \left(\text{since } \hat{T} = \frac{dr}{ds} \right)$$

dot product UNIT tangent vector

Thus, the total work

$$W = \int_C F \cdot \hat{T} ds = \int_C F \cdot dr = \int_C F_1 dx + F_2 dy + F_3 dz = \int_a^b F(r(t)) r'(t) dt$$

In general, if $F = F_1 \vec{i} + F_2 \vec{j} + F_3 \vec{k}$ is a continuous vector field and C is an oriented smooth curve then,

$$\int_C F \cdot dr = \int_C F \cdot \hat{T} ds = \int_C F_1(x, y, z) dx + F_2(x, y, z) dy + F_3(x, y, z) dz \quad (\star)$$

Such integral given $\oint$ is called, somewhat improperly the line integral of F along C (It is not the line integral of F , which should have a vector value, but rather the line integral of the tangential component of F , which has a scalar value)

$\uparrow$ This
 $\circledast$ Line integral depends on the direction of the orientation of the curve C along which the integral is calculated. reversing the direction of C causes this line integral to change sign.
 $\int_C F \cdot dr = - \int_{-C} F \cdot dr$

$\rightarrow$ If C is a closed curve, the line integral of the tangential component of F around C is also called the circulation of F around C . The fact that the curve is closed is often by a small circle drawn on the integral sign.

$\oint_C F \cdot dr$ denotes the circulation of F around the closed curve C .

$\rightarrow$ For a smooth arc $r = r(t) = x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k}$ ($a \leq t \leq b$), we have

$$\int_C F \cdot dr = \int_a^b F \cdot \frac{dr}{dt} dt = \int_a^b [F_1(x,y,z) \frac{dx}{dt} + F_2(x,y,z) \frac{dy}{dt} + F_3(x,y,z) \frac{dz}{dt}] dt$$

Independence of parametrization
 This line integral is independent of the choice of any particular parametrization, used for the curve with a fixed orientation.

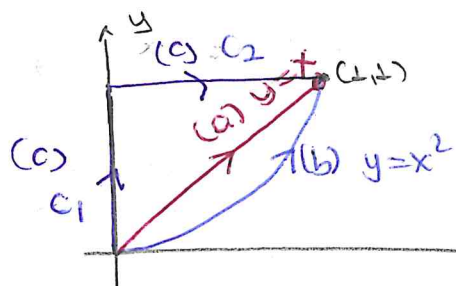
In other words, as long as the curve and its direction remain the same, two parametrizations trace the same path, just at different speeds, the integral gives the same result.

Ex 7 $r_1(t) = t\vec{i} + t^2\vec{j}$ $\int F \cdot dr_1 = \int_0^1 (t\vec{i} + t^2\vec{j}) \cdot (1\vec{i} + 2t\vec{j}) dt = \int_0^1 2t dt = 1$
 $r_2(t) = t^2\vec{i} + t^3\vec{j}$ $\int F \cdot dr_2 = \int_0^1 (t^2\vec{i} + t^3\vec{j}) \cdot (2t\vec{i} + 3t^2\vec{j}) dt = \int_0^1 4t^3 dt = 1$
 Parametrizing the curve $y = x^2$, $0 \leq x \leq 1$ over a piecewise smooth curve

$\rightarrow$ A line integral over the smooth is the sum of the line integrals over the smooth arcs constituting the curve.

Ex 7 $F(x,y) = y^2 \vec{i} + 2xy \vec{j}$ evaluate $\int_C F \cdot d\vec{r}$ from $(0,0)$ to $(1,1)$

- (a) the straight line $y=x$
 (b) the curve $y=x^2$
 (c) the piecewise smooth path consisting of the straight line segments from $(0,0)$ to $(0,1)$ and from $(0,1)$ to $(1,1)$



(a) $y=x$ $\vec{r}(t) = t\vec{i} + t\vec{j}$ $0 \leq t \leq 1$
 $\vec{r}'(t) = \vec{i} + \vec{j}$
 $F \cdot d\vec{r} = (t^2\vec{i} + 2t^2\vec{j}) \cdot (\vec{i} + \vec{j}) dt$
 $= t^2 + 2t^2 = 3t^2 dt$
 $\int_C F \cdot d\vec{r} = \int_0^1 3t^2 dt = t^3 \Big|_0^1 = 1$

(b) $\vec{r}(t) = t\vec{i} + t^3\vec{j}$ $(0 \leq t \leq 1)$ $\vec{r}'(t) = \vec{i} + 3t^2\vec{j}$ $(0 \leq t \leq 1)$
 $F \cdot d\vec{r} = (t^4\vec{i} + 2t^3\vec{j}) \cdot (\vec{i} + 3t^2\vec{j})$
 $= t^4 + 6t^5 = 5t^4 dt$
 $\int_C F \cdot d\vec{r} = \int_0^1 5t^4 dt = t^5 \Big|_0^1 = 1$

(c) $C = C_1 \cup C_2$

$-C_1: x=0$ $\vec{r} = 0\vec{i} + t\vec{j}$ $0 \leq t \leq 1$ $\vec{r}' = \vec{j}$ $d\vec{r} = \vec{j} \cdot dt$
 $F \cdot d\vec{r} = (t^2\vec{i} + 0\vec{j}) \cdot \vec{j} = 0$

$-C_2: y=1$ $x=t$ $0 \leq t \leq 1$ $\vec{r} = t\vec{i} + \vec{j}$ $d\vec{r} = \vec{i} \cdot dt$
 $F \cdot d\vec{r} = (1\vec{i} + 2t\vec{j}) \cdot \vec{i} = 1$

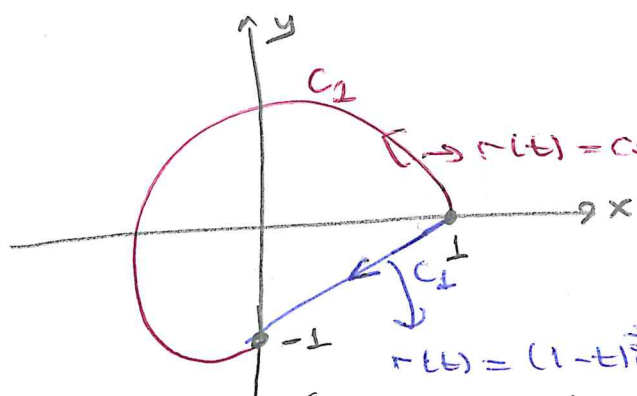
$\int_C F \cdot d\vec{r} = \int_{C_1} F \cdot d\vec{r} + \int_{C_2} F \cdot d\vec{r}$
 $= \int_0^1 0 \cdot dt + \int_0^1 1 \cdot dt = 1$

(a), (b), (c) gives the same result

Ex 7 Let $\vec{F} = y\vec{i} - x\vec{j}$. Find $\int_C \vec{F} \cdot d\vec{r}$ from $(1,0)$ to $(0,-1)$ along

(a) the straight line segment C_1 joining these points

(b) three-quarters of the unit circle centered at the origin and traversed counterclockwise



$$C_1: \vec{r}(t) = (1-t)\vec{i} - t\vec{j} \quad 0 \leq t \leq 1$$

$$d\vec{r} = -dt\vec{i} - dt\vec{j}$$

$$\vec{F}(\vec{r}(t)) = -t\vec{i} - (1-t)\vec{j}$$

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^1 (-t)(-dt) - (1-t)(-dt)$$

$$= \int_0^1 t dt + dt - t dt = \int_0^1 1 dt = 1$$

$$\vec{r}(t) = (1-t)\vec{i} - t\vec{j} \quad 0 \leq t \leq 1$$

Line segment joining two points (x_0, y_0) and (x_1, y_1)

$$x(t) = (1-t)x_0 + tx_1$$

$$y(t) = (1-t)y_0 + ty_1$$

$$(x_0, y_0) = (1, 0) \quad (x_1, y_1) = (0, -1) \quad \vec{r}(t) = (1-t)\vec{i} - t\vec{j}$$

$$x(t) = (1-t) \cdot 1 + t \cdot 0 = 1-t$$

$$y(t) = (1-t) \cdot 0 + t \cdot (-1) = -t$$

$$C_2: \vec{r}(t) = \cos t \vec{i} + \sin t \vec{j}$$

$$\vec{F}(\vec{r}(t)) = \sin t \vec{i} - \cos t \vec{j}$$

$$d\vec{r} = -\sin t dt \vec{i} + \cos t dt \vec{j}$$

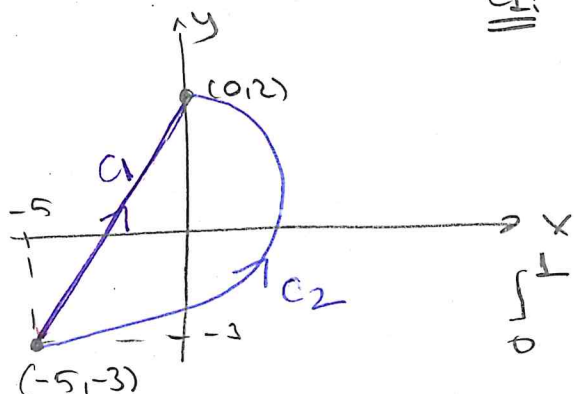
$$\int_C \vec{F} \cdot d\vec{r} = \int_0^{3\pi/2} (\sin t (-\sin t) - \cos t (\cos t)) dt$$

$$= \int_0^{3\pi/2} (-\cos^2 t - \sin^2 t) dt = -\int_0^{3\pi/2} 1 dt = -3\pi/2$$

Ex 7 Evaluate $\int_C y^2 dx + x dy$ where

(a) $C = C_1$ line segment from $(-5, -3)$ to $(0, 2)$

(b) $C = C_2$ the arc of the parabola $x = 4 - y^2$ from $(-5, -3)$ to $(0, 2)$



$$C_1: x(t) = (1-t)(-5) + t(0) \quad (x_0, y_0) = (-5, -3)$$

$$y(t) = (1-t)(-3) + t(2) \quad (x_1, y_1) = (0, 2)$$

$$\vec{r}(t) = (5t - 5)\vec{i} + (5t - 3)\vec{j}$$

$$d\vec{r} = 5 dt \vec{i} + 5 dt \vec{j}$$

$$\int_0^1 (5t-3)^2 \cdot 5 dt + (5t-5) \cdot 5 dt = \sqrt{-5/6}$$

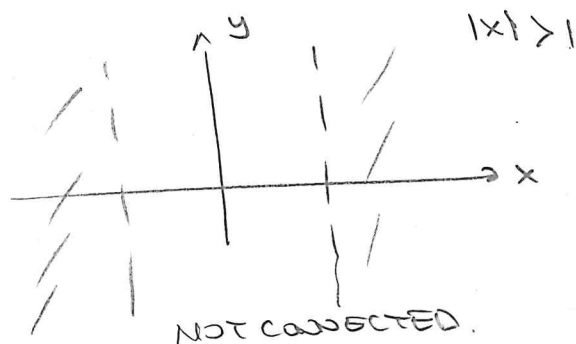
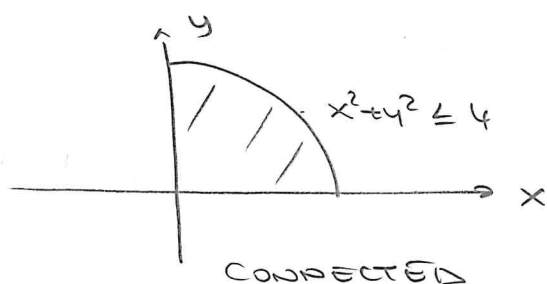
⑥ $x = 4 - y^2$ $y = t$ $x = 4 - t^2$ $-3 \leq t \leq 2$
 $dy = dt$ $dx = -2t dt$

$$\int_C y^2 dx + x dy = \int_{-3}^2 t^2 (-2t) + (4 - t^2) \cdot 1 dt = \left[-\frac{t^4}{2} - \frac{t^3}{3} + 4t \right]_{-3}^2$$

$$= \left[\left(-\frac{16}{2} - \frac{8}{3} + 8 \right) - \left(-\frac{81}{2} + \frac{27}{3} - 12 \right) \right] = \frac{245}{6}$$

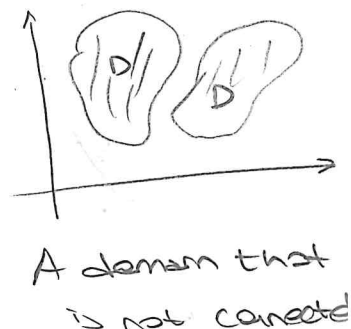
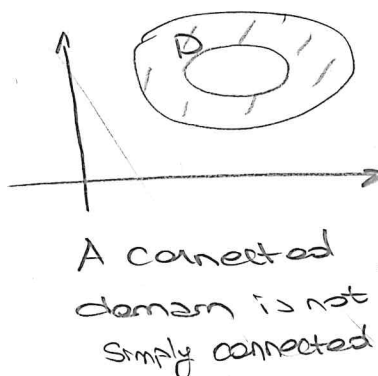
Connected and simply Connected Domains

Definition A domain D is to be **connected** if every pair of points P and Q in D can be joined by a piecewise smooth curve lying in D



→ A closed curve is **simple** if it has no self-intersections other than beginning and ending at the same point (For example, a circle is a simple closed curve)

→ A **simply connected domain** D is a connected domain in which every simple closed curve can be continuously shrunk to a point, locally in D without any part ever passing out of D .



→ In the plane, a simply connected domain D , can have no holes, not even a hole consisting of a single point

→ In 3-space, a simply connected domain can have holes.

→ The set of all points in $\mathbb{R}^3$ different from the origin ($\mathbb{R}^3 \setminus \{0\}$) is simply connected. As is the exterior of a ball that is $\{(x,y,z) : \sqrt{x^2+y^2+z^2} > R\}$

→ But the set of all points in $\mathbb{R}^3$ satisfying $x^2+y^2 > 0$ is not simply connected. In other words, $\{(x,y,z) \in \mathbb{R}^3 \mid x^2+y^2 > 0\}$ is not simply connected. z -axis acts like an infinitely long "hole".

Summary. $\mathbb{R}^3 \setminus \{0\}$ is simply connected, removing a small point

doesn't stop you from making a rope loop around

it and then tightening it - you can go over, under or around

Consider $\gamma(t) = (r \cos t, r \sin t, 0)$ $t \in [0, 2\pi]$. let $s \in [0, 1]$ $\gamma_s(t) = ((1-s)r \cos t, (1-s)r \sin t, s)$

As $s \rightarrow 1$ radius $\rightarrow 0$ Height $\rightarrow 1$ loop shrinks to $(0,0,1)$ without touching origin

• $\mathbb{R}^3 \setminus \{z\text{-axis}\}$ not simply connected, because the missing set is

1-dimensional and infinite, making it act like a "strip"

piercing space.
down

• $\mathbb{R}^2 \setminus \{0\}$ is not simply connected. A circle around the origin

in the plane cannot shrink to a point "without crossing the origin" there is no "above" or "below" to escape to.

THEM INDEPENDENCE OF PATH (FUNDAMENTAL THEOREM OF LINE INTEGRALS)

Let D be an open connected domain, and let F be a smooth vector field defined on D . Then, the following three statements are equivalent in the sense that, if any one of them is true, so are the other two:

- ① F is conservative in D .
- ② $\oint_C F \cdot dr = 0$ for every piecewise smooth, closed curve C in D .
- ③ Given any two points P_0 and P_1 in D , $\int_C F \cdot dr$ has the same value for all piecewise smooth curves in D

Starting at P_0 and ending at P_1 .

REMARK It is very easy to evaluate the line integral of the tangential component of a conservative vector field along a curve C , when you know a potential for F . If $F = \nabla \phi$, and C goes from P_0 to P_1 then

$$\int_C F \cdot dr = \int_C d\phi = \phi(P_1) - \phi(P_0)$$

Because

$$F \cdot dr = \left(\frac{\partial \phi}{\partial x} \vec{i} + \frac{\partial \phi}{\partial y} \vec{j} + \frac{\partial \phi}{\partial z} \vec{k} \right) \cdot (dx \vec{i} + dy \vec{j} + dz \vec{k})$$
$$= \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz = d\phi$$

and so

$$\int_C F \cdot dr = \int_a^b \frac{d\phi(r(t))}{dt} dt = \phi(\underbrace{r(b)}_{P_1}) - \phi(\underbrace{r(a)}_{P_0})$$

Ex 7 For what values of the constants A and B is the vector field

$$F = Ax \sin(\pi y) \vec{i} + (x^2 \cos(\pi y) + By e^{-z}) \vec{j} + y^2 e^{-z} \vec{k}$$

conservative? For this choice of A and B evaluate $\int_C F \cdot dr$

(a) C is the curve $r = \cos t \vec{i} + \sin 2t \vec{j} + \sin^2 t \vec{k}$ ($0 \leq t \leq 2\pi$) and

(b) the curve of intersection of the paraboloid $z = x^2 + 4y^2$ and the plane $z = 3x - 2y$ from $(0,0,0)$ to $(1, 1/2, 2)$

(a) Conservative: $\frac{\partial F_1}{\partial y} = \frac{\partial F_2}{\partial x}$ $\frac{\partial F_1}{\partial z} = \frac{\partial F_3}{\partial x}$ $\frac{\partial F_2}{\partial z} = \frac{\partial F_3}{\partial y}$

$$A \pi x \cos(\pi y) = 2x \cos(\pi y) \quad 0 = 0 \quad -By e^{-z} = 2y e^{-z}$$

We need $A = 2/\pi$ and $B = -2$.

$F = \nabla \phi$ where $\phi(x,y,z) = \frac{x^2 \sin(\pi y)}{\pi} - y^2 e^{-z} + C$

$$\left[\begin{aligned} \phi &= \int \frac{2x}{\pi} \sin(\pi y) dx + h(y,z) & \frac{\partial \phi}{\partial z} &= h_z(y,z) = -y^2 e^{-z} \\ &= \frac{x^2}{\pi} \sin(\pi y) + h(y,z) & h(y,z) &= -y^2 e^{-z} + g(y) \\ \phi(x,y,z) &= \frac{x^2 \sin(\pi y)}{\pi} - y^2 e^{-z} + g(y) & \frac{\partial \phi}{\partial y} &= \frac{x^2 \cos(\pi y) \cdot \pi}{\pi} - 2y e^{-z} = g'(y) \\ & & g'(y) &= 0 \quad g(y) = C \end{aligned} \right]$$

$$\phi(x,y,z) = \frac{x^2 \sin(\pi y)}{\pi} - y^2 e^{-z} + C$$

(a) $r = \cos t \vec{i} + \sin 2t \vec{j} + \sin^2 t \vec{k} \quad 0 \leq t \leq 2\pi$

$r(0) = \vec{i} = r(2\pi)$ closed curve

$$\int_C F \cdot dr = \oint_C \nabla \phi \cdot dr = \phi(r(0)) - \phi(r(2\pi)) = 0$$

(b) $\int_C F \cdot dr = \left(\frac{x^2 \sin(\pi y)}{\pi} - y^2 e^{-z} \right) \Big|_{(0,0,0)}^{(1, 1/2, 2)} = \frac{1}{\pi} - \frac{1}{4e^2}$

Ex Evaluate
$$I = \oint_C (e^x \sin y + 3y) dx + (e^x \cos y + 2x - 2y) dy$$

where C is counterclockwise ellipse $4x^2 + y^2 = 4$

$$F = \underbrace{(e^x \sin y + 3y)}_{F_1} \vec{i} + \underbrace{(e^x \cos y + 2x - 2y)}_{F_2} \vec{j} \rightarrow \text{not conservative.}$$

$$\frac{\partial F_1}{\partial y} = e^x \cos y + 3 \neq \frac{\partial F_2}{\partial x} = e^x \cos y + 2.$$

Consider $\phi(x, y) = e^x \sin y + 2xy - y^2$. $F = \nabla \phi + y \vec{j}$

$$I = \underbrace{\oint_C \nabla \phi \cdot d\vec{r}}_{I_1} + \underbrace{\oint_C y dx}_{I_2}$$

$I_1 =$ This integral is zero because of conservative field and C is closed

ellipse $4x^2 + y^2 = 4$ $x = \cos t$ $y = 2 \sin t$ ($0 \leq t \leq 2\pi$) $dx = -\sin t dt$
 $dy = 2 \cos t dt$

$$I_2 = \oint_C y \cdot dx = - \oint_C 2 \sin t (-\sin t) dt = -2 \int_0^{2\pi} \sin^2 t dt$$

$$= -2 \int_0^{2\pi} \frac{1 - \cos(2t)}{2} dt = -2\pi$$

$$I = -2\pi.$$

Ex 7. Evaluate $\int_C \vec{\nabla} f \cdot d\vec{r}$ where $f(x,y,z) = \cos(\pi x) + \sin(\pi y) - x^2 z$.

where C is any path starts at $(1, 1/2, 2)$ and ends at $(2, 1, -1)$.

Let $\vec{r}(t)$, $a \leq t \leq b$, be any path that starts at $(1, 1/2, 2)$ and ends at $(2, 1, -1)$ so $\vec{r}(a) = (1, 1/2, 2)$ $\vec{r}(b) = (2, 1, -1)$

$$\int_C \vec{\nabla} f \cdot d\vec{r} = f(\vec{r}(b)) - f(\vec{r}(a)) = [\cos(2\pi) + \sin(\pi) - 2 \cdot 1 \cdot (-1)] - [\cos(\pi) + \sin(\pi/2) - 1 \cdot 1/2 \cdot 2] = (1 + 0 + 2) - (-1 + 1 + 1) = \sqrt{4}$$

Summary

$F = \vec{\nabla} f$ on $D \Leftrightarrow F$ conservative on $D \Leftrightarrow \oint_C \vec{F} \cdot d\vec{r} = 0$ over any closed path

$\Leftrightarrow \int_C \vec{F} \cdot d\vec{r} = F(\vec{r}(a)) - F(\vec{r}(b))$ $\vec{r}(t)$, $a \leq t \leq b$ any curve representing C , connecting points $\vec{r}(a)$ and $\vec{r}(b)$

Ex 7 (a) $F(x,y) = (3+2xy)\vec{i} + (x^2-3y^2)\vec{j}$ Find $F = \vec{\nabla} f$

(b) Evaluate line integral $\int_C \vec{F} \cdot d\vec{r}$ where C is the curve given by $\vec{r}(t) = e^t \sin t \vec{i} + e^t \cos t \vec{j}$ $0 \leq t \leq \pi$

(a) $f_x = 3 + 2xy$
 $f_y = x^2 - 3y^2$

$f(x,y) = \int (3 + 2xy) dx + h(y)$
 $= x^2 y + h(y)$
 $f_y = x^2 + h'(y) = x^2 - 3y^2$ $h'(y) = -3y^2$
 $h(y) = -y^3 + C$

$f(x,y) = x^2 y - y^3 + C$

(b) $\int_A^B \vec{F} \cdot d\vec{r} = f(B) - f(A)$
 $A(0, -e^\pi)$

$\int_{(0,1)}^{(0,-1)} \vec{F} \cdot d\vec{r} = f(0, -e^\pi) - f(0, 1)$
 $= e^{3\pi} + 1$

$\vec{r}(0) = e^0 \sin 0 \vec{i} + e^0 \cos 0 \vec{j} = (0, 1)$
 $\vec{r}(\pi) = e^\pi \sin \pi \vec{i} + e^\pi \cos \pi \vec{j} = (0, -e^\pi)$

GREEN'S THEOREM IN THE PLANE

→ Fundamental theorem of calculus

$$\int_a^b \frac{d}{dx} f(x) dx = f(b) - f(a)$$

express the integral, taken over the interval $[a, b]$, of the derivative of a single variable function, f , as a sum of values of the function at the oriented boundary of the interval $[a, b]$.

→ The line integral of a conservative vector field over a curve C from A to B

$$\int_C \nabla \phi \cdot d\mathbf{r} = \phi(B) - \phi(A)$$

has similar interpretation; $\nabla \phi$ is a derivative, and the curve C , although lying in a two- or three dimensional space, it is intrinsically a one-dimensional object; (you need one parameter to express it which is t , coming from $\mathbf{r}(t)$) and the points A and B constitute its boundary

→ Green's Theorem is a two-dimensional version of the Fundamental Theorem of Calculus that expresses double integral of a certain kind of derivative of a two dimensional vector field $F(x, y)$, namely, the k -th component of $\text{Curl } F$, over a region R in the xy -plane as a line integral of the tangential component of F around the curve C , which is oriented boundary of R :

$$\text{If } F(x, y) = F_1(x, y)\vec{i} + F_2(x, y)\vec{j} \quad \text{Curl } F = \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \vec{k} \quad (\text{see page 5})$$

$$\iint_R \text{Curl } F \cdot \vec{k} \, dA = \oint_C F \cdot d\mathbf{r}$$

More explicitly

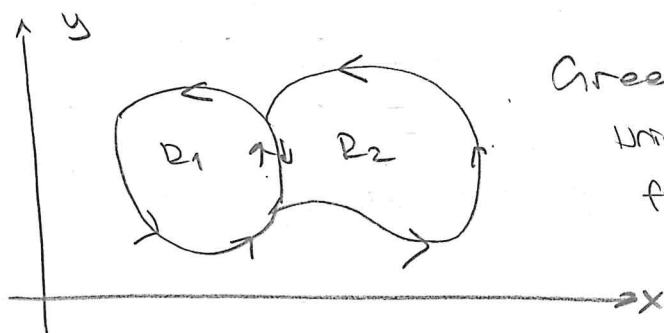
$$\iint_R \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dx dy = \oint_C F_1(x, y) dx + F_2(x, y) dy$$

THOM GREEN'S THEOREM

Let R be a regular, closed in the xy -plane whose boundary, C , consists of one or more piecewise smooth, simple closed curves that are positively oriented (counterclockwise) with respect to R . If $\vec{F} = F_1(x,y)\vec{i} + F_2(x,y)\vec{j}$ is a smooth vector field on R , then

$$\oint_C \vec{F} \cdot d\vec{r} = \oint_C F_1(x,y)dx + F_2(x,y)dy = \iint_R \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dA.$$

Counter-clockwise circulation
Curl integral



Green's Theorem holds for the union of R_1 and R_2 if it holds for each of these regions.

Ex 7 For any of these three vector fields

$$\vec{F}^1 = x\vec{j} \quad F_1^1 = 0 \quad F_2^1 = x \quad \frac{\partial F_2^1}{\partial x} - \frac{\partial F_1^1}{\partial y} = 1.$$

$$\vec{F}^2 = -y\vec{i} \quad F_1^2 = -y \quad F_2^2 = 0 \quad \frac{\partial F_2^2}{\partial x} - \frac{\partial F_1^2}{\partial y} = 1.$$

$$\vec{F}^3 = \frac{1}{2}(-y\vec{i} + x\vec{j}) \quad F_1^3 = -\frac{y}{2} \quad F_2^3 = \frac{x}{2} \quad \frac{\partial F_2^3}{\partial x} - \frac{\partial F_1^3}{\partial y} = 1$$

If C is positively oriented piecewise smooth, simple closed curve bounding the region R in the plane Green's theorem gives

$$\underbrace{\oint_C x dy}_{F_1} = - \underbrace{\oint_C y dx}_{F_2} = \frac{1}{2} \underbrace{\oint_C x dy - y dx}_{F_3} = \iint_R 1 dA = \text{area of } R$$

Ex 7 Using previous rule calculate the area of the ellipse
disk bounded by the curve C given by

$$r(t) = 3(\cos t + \sin t)\vec{i} + 2(\sin t - \cos t)\vec{j} \quad 0 \leq t \leq 2\pi$$

$$x(t) = 3(\cos t + \sin t)$$

$$y(t) = 2(\sin t - \cos t)$$

$$dx = 3(-\sin t + \cos t) dt$$

$$dy = 2(\cos t + \sin t) dt$$

- Consider $F = \frac{1}{2}(-y\vec{i} + x\vec{j})$ then we have

$$x dy - y dx = 6((\cos t + \sin t)^2 + (\sin t - \cos t)^2) dt = 12 dt$$

$$\text{Area} = \frac{1}{2} \oint_C x dy - y dx = \frac{1}{2} \int_0^{2\pi} 12 dt = \underline{12\pi}$$

- Consider $F = x\vec{j}$ then we have

$$x dy = \underbrace{3(\cos t + \sin t)}_x \cdot \underbrace{2(\sin t + \cos t) dt}_{dy}$$

$$= 6(\cos t + \sin t)^2 dt$$

$$\text{Area} = \int_C x dy = \int_0^{2\pi} 6(\cos t + \sin t)^2 dt =$$

$$\int_0^{2\pi} 6(\cos^2 t + 2\cos t \sin t + \sin^2 t) dt = \int_0^{2\pi} 6(1 + \underbrace{2\cos t \sin t}_{\sin 2t}) dt$$

$$= (6t - 6\cos 2t) \Big|_0^{2\pi} = 12\pi - 6\cos 4\pi - (0 - 6\cos 0) \\ = 12\pi - 6 + 6 = \underline{12\pi}$$

- Consider $F = -y\vec{j}$

$$-y dx = -2(\sin t - \cos t) \cdot 3(-\sin t + \cos t) dt \\ = 6(\sin t - \cos t)^2$$

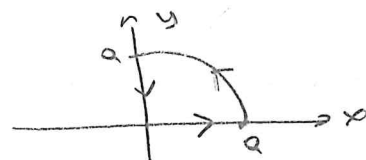
$$\int_C -y dx = \int_0^{2\pi} 6(\sin t - \cos t)^2 dt = \int_0^{2\pi} 6(1 - 2\underbrace{\cos t \sin t}_{\sin 2t}) dt$$

$$= (6t + 6\cos 2t) \Big|_0^{2\pi} = 12\pi + 6\cos 4\pi - (0 + 6\cos 0) \\ = \underline{12\pi}$$

(27)

Ex 7 Evaluate $I = \oint_C (x-y^3)dx + (y^3+x^3)dy$ where C is the positively oriented boundary of the quarter-disk Θ :

$$0 \leq x^2 + y^2 \leq a^2, \quad x \geq 0, \quad y \geq 0.$$

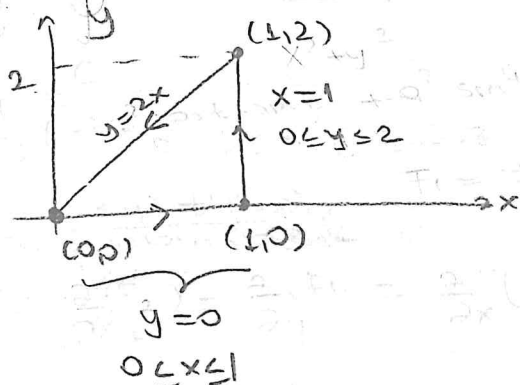


$$I = \iint_{\Theta} \left(\frac{\partial}{\partial x} (y^3 + x^3) - \frac{\partial}{\partial y} (x - y^3) \right) dA$$

$$= \iint_{\Theta} (3x^2 + 3y^2) dA = 3 \int_0^{\pi/2} \int_0^a r^2 \cdot r dr = \frac{3}{8} \pi a^4$$

Ex 7 Use Green's theorem to evaluate $\oint_C xy dx + x^2 y^3 dy$ where C is the triangle with vertices $(0,0)$, $(1,0)$, $(1,2)$ with positive orientation.

Positive orientation.



Green's Theorem

C : simple, closed, piecewise smooth curve

$$P = xy \quad Q = x^2 y^3$$

polynomials with continuous derivatives

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 2xy^3 - x$$

$$\oint_C xy dx + x^2 y^3 dy = \int_0^1 \int_0^{2x} (2xy^3 - x) dy dx = \int_0^1 \left(\frac{xy^4}{2} - xy \right) \Big|_0^{2x} dx$$

$$= \int_0^1 \left(\frac{16x^5}{2} - 2x^2 \right) dx = \frac{8x^6}{6} - \frac{2x^3}{3} \Big|_0^1 = \frac{2}{3}$$

Line Integral

$$C_1: x=t \quad y=0$$

$$dx=dt \quad dy=0$$

$$0 \leq t \leq 1 \quad \int_0^1 0+0=0$$

C_1

$$C = C_1 \cup C_2 \cup C_3$$

$$\oint_C = \int_{C_1} + \int_{C_2} + \int_{C_3} = 0 + 4 - \frac{10}{3} = \frac{2}{3}$$

$$C_2: x=1 \quad y=t \quad 0 \leq t \leq 2$$

$$dx=0 \quad dy=dt$$

$$\int_{C_2} t^3 dt = \int_0^2 t^3 dt = \frac{t^4}{4} \Big|_0^2 = 4$$

$$C_3: x=t \quad dx=dt \quad 1 \leq t \leq 0$$

$$y=2t \quad dy=2dt$$

OR we can use

$$x=1-t \quad dx=-dt \quad 0 \leq t \leq 1$$

$$dy=2-2t \quad dy=-2dt$$

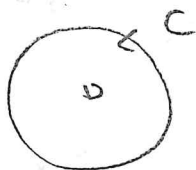
$$\int_{C_3} xy dx + x^2 y^3 dy = \int_1^0 (2t^2 + 16t^5) dt$$

$$= \frac{2t^3}{3} + \frac{16t^6}{6} \Big|_1^0 =$$

$$= -\frac{2}{3} - \frac{8}{3} = -\frac{10}{3}$$

Ex 7 Evaluate $\oint_C (3y - e^{\sin x}) dx + (7x + \sqrt{y^4 + 1}) dy$ where

C is the positively oriented circle with $x^2 + y^2 = 9$



C : closed, simple, smooth curve

$$F_1 = 3y - e^{\sin x}$$

have continuous partial

$$F_2 = 7x + \sqrt{y^4 + 1}$$

$$\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} = (7 - 3) = 4$$

derivatives

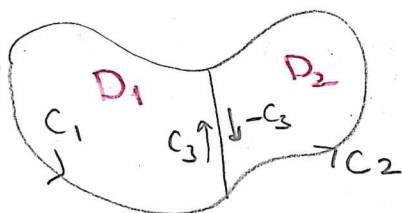
$$\oint_C (3y - e^{\sin x}) dx + (7x + \sqrt{y^4 + 1}) dy = \iint_D 4 dA = \int_0^{2\pi} \int_0^3 4r dr d\theta = 36\pi$$

REMARKS ① If we know $F_1(x,y) = F_2(x,y) = 0$ on the curve C

$$\iint_R \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dA = \oint_C F_1 dx + F_2 dy = 0 \text{ no matter}$$

what values $F_1(x,y)$ and $F_2(x,y)$ in the region

② We prove Green's theorem for D is simple, we can extend it to the case where D is a finite union of simple regions



$D = D_1 \cup D_2$ Boundary of D_1 is $C_1 \cup C_3$

Boundary of D_2 is $C_2 \cup (-C_3)$

$$C = C_1 \cup C_2$$

We can apply Green's Theorem D_1 and D_2 separately

$$\oint_{C_1 \cup C_3} F_1 dx + F_2 dy = \iint_{D_1} \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dA$$

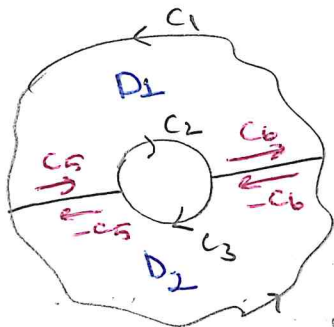
$$\oint_{C_2 \cup (-C_3)} F_1 dx + F_2 dy = \iint_{D_2} \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dA$$

Add.

$$\oint_{C_1 \cup C_3} F_1 dx + F_2 dy + \oint_{C_2 \cup (-C_3)} F_1 dx + F_2 dy = \int_{C_1} F_1 dx + F_2 dy + \int_{C_3} F_1 dx + F_2 dy$$

$$+ \int_{C_2} F_1 dx + F_2 dy - \int_{C_3} F_1 dx + F_2 dy = \oint_{C_1 \cup C_2} F_1 dx + F_2 dy = \iint_{D_1 \cup D_2} \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dA$$

Similarly, we can use.

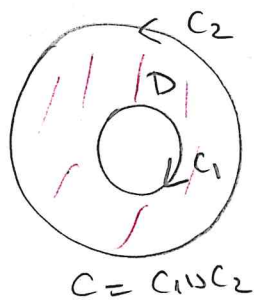


We have

$$\iint_{D=D_1 \cup D_2} ((F_2)_x - (F_1)_y) dA = \oint_{C_1 \cup C_2 \cup C_3 \cup C_4} F_1 dx + F_2 dy$$

Please note that C_1 and C_4 are positively oriented while C_2 and C_3 are negatively oriented

Ex 7 Evaluate $\oint_C y^3 dx - x^3 dy$ where C are two circles of radius 2 and radius 1 centered at the origin with positive and negative orientation, respectively



$$C = C_1 \cup C_2$$

$$\oint_C y^3 dx - x^3 dy = \iint_D (-3x^2 - 3y^2) dA = -3 \int_0^{2\pi} \int_1^2 r^2 \cdot r dr d\theta = -45 \frac{\pi}{2}$$

$$F_1 = y^3 \quad F_2 = -x^3$$

$$\frac{\partial F_1}{\partial y} = 3y^2 \quad \frac{\partial F_2}{\partial x} = -3x^2$$

Ex 7. Let C be a positively oriented, simple closed curve in the xy -plane bounding a region R and not passing through the origin.

Show that

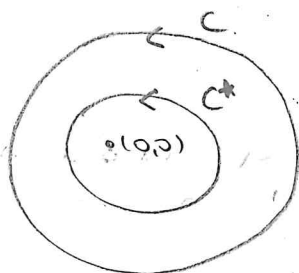
$$\oint_C \frac{-y dx + x dy}{x^2 + y^2} = \begin{cases} 0 & \text{if the origin is outside } R \\ 2\pi & \text{if the origin is inside } R. \end{cases}$$



R not involving origin so $\frac{\partial}{\partial x} \left(\frac{x}{x^2+y^2} \right) - \frac{\partial}{\partial y} \left(\frac{-y}{x^2+y^2} \right) = 0$
 $(x,y) \neq (0,0)$

$$\oint_C \frac{-y dx + x dy}{x^2 + y^2} = \iint_R \left[\frac{\partial}{\partial x} \left(\frac{x}{x^2+y^2} \right) - \frac{\partial}{\partial y} \left(\frac{-y}{x^2+y^2} \right) \right] dx dy = 0$$

R involving origin



Let us consider a counter-clockwise-oriented circle C^* with center origin and radius a as in the figure

★ Caves $\Rightarrow \int_{C \cup C^*} F_1 dx + F_2 dy = \iint_D ((F_2)_x - (F_1)_y) dA$

$$= \iint_D \left(\frac{y^2 - x^2}{(x^2 + y^2)^2} - \frac{y^2 - x^2}{(x^2 + y^2)^2} \right) dA = 0$$

Therefore, $\oint_{C \cup C^*} F_1 dx + F_2 dy = \oint_C F_1 dx + F_2 dy + \oint_{-C^*} F_1 dx + 0 dy = 0$
 $\left(- \int_{C^*} F_1 dx + 0 dy \right)$

$$\oint_C F_1 dx + F_2 dy = \int_{C^*} F_1 dx + F_2 dy$$

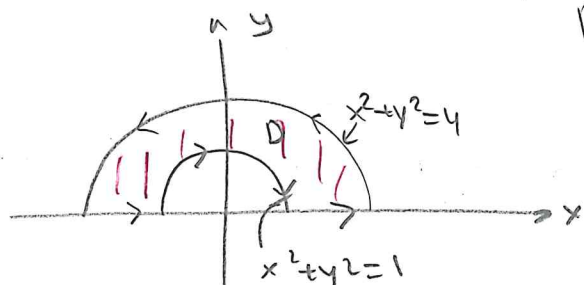
Parametrization of C^* $\vec{r}(t) = a \cos t \vec{i} + a \sin t \vec{j}$ $0 \leq t \leq 2\pi$
 $dx = -a \sin t dt$ $dy = a \cos t dt$

$$\int_0^{2\pi} \frac{-a \sin t (-a \sin t) dt + a \cos t (a \cos t) dt}{a^2} = \int_0^{2\pi} 1 dt = 2\pi$$

$$\oint_C \frac{-y dx + x dy}{x^2 + y^2} = \oint_{C^*} \frac{-y dx + x dy}{x^2 + y^2} = 2\pi$$

(31)

Ex 7 Evaluate $\oint_C y^2 dx + 3xy dy$ where C is the boundary of the semiannular region D in the upper half-plane between the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$



$$D = \{ (r, \theta) \mid 1 \leq r \leq 2 \quad 0 \leq \theta \leq \pi \}$$

$$\begin{aligned} \oint_C y^2 dx + 3xy dy &= \iint_D \left[\frac{\partial}{\partial x} (3xy) - \frac{\partial}{\partial y} (y^2) \right] dA \\ &= \iint_D y dA = \int_0^\pi \int_1^2 (r \sin \theta) r dr d\theta \\ &= \int_0^\pi \int_1^2 r^2 \sin \theta dr d\theta = \boxed{\frac{14}{3}} \end{aligned}$$

REMARK 3 $\int_C \vec{F} \cdot d\vec{r}$ is path independent in $D \Leftrightarrow \oint_C \vec{F} \cdot d\vec{r} = 0$ for every closed path C in D
(Page 22 Thm 1, 2, 3)

$\Rightarrow$ If $\int_C \vec{F} \cdot d\vec{r}$ is path independent in D , then $\vec{F} = F_1 \vec{i} + F_2 \vec{j}$

conservative vector field that is $\frac{\partial F_1}{\partial y} = \frac{\partial F_2}{\partial x}$

$$\oint_C \vec{F} \cdot d\vec{r} = \underbrace{\iint_D \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) dA}_{\text{Green's theorem}} = \iint_D 0 dA = 0$$